Week 22 — Extended Reading: Constrained objectives and practical updates

STUDY-RL

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Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Ray, Achiam and Amodei (2019): Safety Gym

Primary source. https://cdn.openai.com/safexp-short.pdf

Mechanism and scope. The paper introduces a benchmark for safe exploration and compares constrained-learning approaches using reward and cost measures. It supplies an evaluation framework, not a proof that a neural Lagrangian policy never violates a constraint.

Read, derive, investigate. Distinguish final expected cost from costs incurred while learning. Which reported quantity counts constraint violations before convergence? The course lava grid uses a simpler state/action space, so its runtime and curves are not benchmark reproductions.

2 Achiam et al. (2017): CPO

Primary source. https://arxiv.org/abs/1705.10528

Mechanism and scope. CPO derives policy-performance bounds and a practical constrained trust-region update. Local linear/quadratic approximations, sampled gradients and approximate line searches add implementation limitations beyond the ideal inequalities.

Read, derive, investigate. For the quadratic subproblem, derive the step $H^{-1}(g-\nu b)/\lambda$ and the sign of each dual term. Check feasibility before using the formula. If $c > \sqrt{2\delta b^\top H^{-1} b}$, explain why a recovery step reduces cost without satisfying the original linearised constraint in one step.

3 Zhang, Vuong and Ross (2020): FOCOPS

Primary source. https://arxiv.org/abs/2002.06506

Mechanism and scope. FOCOPS derives a nonparametric exponential-tilt target, then projects using $D_{\mathrm{KL}}(\pi_\theta \| \pi^\star)$. The ideal multipliers are coupled. The practical method uses a fixed temperature and an adaptive cost multiplier with KL controls.

Read, derive, investigate. Read the projection direction directly from Eq. 10. Reversing it yields another optimisation problem. Restore the $(1-\gamma)$ factor when converting an unnormalised discounted cost budget to normalised-occupancy advantage expectations. The workbook computes a target row, not the entire FOCOPS algorithm.

4 Zhang et al. (2022): P3O

Primary source. https://arxiv.org/abs/2205.11814

Mechanism and scope. P3O applies an exact-penalty perspective to proximal policy optimisation. A reward objective to maximise subtracts a nonnegative violation penalty; a loss to minimise adds it after negating reward.

Read, derive, investigate. Using an optimal multiplier $\lambda^\star$, prove that $\kappa > \lambda^\star$ makes a violating policy strictly worse than a feasible optimum in the ideal globally optimised problem. Which step fails for an arbitrary local neural surrogate optimum? Check the hinge subgradient separately from policy-gradient estimation.

5 Stooke, Achiam and Abbeel (2020): PID Lagrangian

Primary source. https://arxiv.org/abs/2007.03964

Mechanism and scope. PID Lagrangian methods view multiplier adaptation as feedback control, using proportional and derivative information alongside accumulated violation. Their experiments examine response and oscillation under particular systems.

Read, derive, investigate. Plot cost error, accumulated error and its change before combining them. Increasing a coefficient does not universally stabilise a noisy delayed system. When rescaling costs, also rescale the budget and multiplier units; otherwise the control target itself changes.

Connect the reading to your workbook

Use the exact one-state budget example to verify signs before neural training. Derive the empirical lower-tail CVaR with fractional mass at the quantile, then contrast it with upper-tail cost risk. Neither an expected budget nor a lower-tail cost summary is a zero-violation safety certificate.
Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.

Statistics reading bridge

An average-cost budget is different from a tail restriction. Read expectation and variance with the cost distribution, then distinguish mean, maximum and tail quantities explicitly. Keep CMDP/CPO primary references for constrained-optimization results and the precise conditions behind them.

Checked textbook locations: Siegrist § § 4.1, 4.3, PDF pp.439, 460. https://stats.libretexts.org/@go/page/10158. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.